BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 199 · Trees and binary search trees

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. In chain A→B→C, which node follows B?

Q2. What does LIFO mean?

Q3. What must be true before binary search on an array?

Q4. Insert X between A and B: what should new X.next point to?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

root — Topmost tree node.

child — Node directly below a parent.

subtree — A node and all its descendants.

binary search tree — A binary tree ordered by values, with smaller keys to the left and greater to the right.

balanced — Height kept relatively small compared with node count.

New learning

Start at root — Compare the new value 25 with root 40; it belongs in the left subtree.

Compare again — 25 is greater than 20, so move right. The right child currently stores 30.

Insert without breaking the property — 25 is smaller than 30; insert it as the left child of 30.

Worked example

Insert 12 into BST with root 20, left 10 and right 30.

1. Predict — Compare 12 with 20 → go left to 10.

2. Confirm and explain — 12 > 10, so 12 becomes the right child of 10.

We Do · Practise with guidance

Insert 35 into BST with root 25, left 15, right 40.

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35 > 25 → RIGHT
35 < 40 → LEFT
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Predict

35 is greater than 25, so go right.

Confirm and explain

35 is smaller than 40, so insert it as the left child of 40.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

BST root 50, left child 25, right child 75: where does 60 go?

Question 2

Insert 15, 10, 20, 18 into an empty BST. Identify the parent of 18.

Question 3

Explain why a BST can have O(n) worst-case search time despite O(log n) average search on a reasonably balanced tree.

Exit Ticket · Answer before checking

Every binary search tree has logarithmic worst-case lookup time.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. C.

Q2. Last In, First Out.

Q3. The array must be sorted.

Q4. B, before changing A.next to X.

You Do pages · Independent questions

Q1. Left child of 75 because 60>50 and 60<75.

Q2. 18 is the left child of 20, which is the right child of 15.

Q3. A sequence of sorted inserts can make it a chain of n nodes; each lookup may visit every node.

Exit Ticket · Model answer

False. An unbalanced BST may degenerate into a chain with O(n) lookup.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: